

## POCATELLO, ID, USA

WMO: 725780

Lat: 42.920N

Lon: 112.571W

Elev: 4452

StdP: 12.48

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24156

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -0.5       | 4.8 | -6.1                          | 4.7 | 0.4  | -0.6 | 6.3 | 7.3  | 33.7                  | 37.0 | 29.5 | 36.1 | 3.5                   | 20   | 0.664 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 34.8                   | 94.9            | 61.5 | 92.0 | 60.8 | 89.1 | 59.9 | 65.3                | 86.5 | 63.5 | 84.7 | 62.0 | 83.5 | 10.1                 | 260  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 58.8                            | 87.4 | 70.7 | 55.9 | 78.6 | 70.2 | 53.5 | 71.9 | 69.5 | 32.4          | 85.7 | 31.0 | 84.3 | 29.7 | 83.2 |                   |
|                                 |      |      |      |      |      |      |      |      |               |      |      |      |      |      |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| 28.7              | 25.7 | 22.9 | -7.6                       | 99.3 | 6.4                | 2.0 | -12.2                                              | 100.7 | -15.9      | 101.9 | -19.5      | 103.1 | -24.1      | 104.5 |    |    |
|                   |      |      | -8.1                       | 69.3 | 6.2                | 2.7 | -12.6                                              | 71.2  | -16.3      | 72.8  | -19.8      | 74.3  | -24.3      | 76.2  |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 47.4   | 26.1  | 29.8  | 39.1  | 45.7  | 54.2  | 62.8  | 71.6  | 69.5  | 60.1  | 47.2  | 35.7  | 26.1 |
|                                                                                  | DBStd       | 17.50  | 9.79  | 9.25  | 7.50  | 6.94  | 7.29  | 6.90  | 4.82  | 5.00  | 7.14  | 8.00  | 8.84  | 9.92 |
|                                                                                  | HDD50       | 3187   | 741   | 565   | 343   | 163   | 40    | 4     | 0     | 0     | 10    | 147   | 432   | 742  |
|                                                                                  | HDD65       | 6866   | 1205  | 985   | 801   | 578   | 339   | 119   | 9     | 18    | 176   | 552   | 878   | 1206 |
|                                                                                  | CDD50       | 2249   | 0     | 0     | 7     | 35    | 172   | 386   | 670   | 604   | 312   | 60    | 3     | 0    |
|                                                                                  | CDD65       | 458    | 0     | 0     | 0     | 0     | 6     | 51    | 214   | 158   | 28    | 1     | 0     | 0    |
|                                                                                  | CDH74       | 8905   | 0     | 0     | 0     | 27    | 266   | 1220  | 3545  | 2916  | 880   | 51    | 0     | 0    |
|                                                                                  | CDH80       | 4038   | 0     | 0     | 0     | 3     | 63    | 471   | 1794  | 1400  | 298   | 9     | 0     | 0    |
| Wind                                                                             | WSAvg       | 9.4    | 9.1   | 9.8   | 10.7  | 11.5  | 10.1  | 9.7   | 8.3   | 8.2   | 8.1   | 9.2   | 9.3   | 9.2  |
| Precipitation                                                                    | PrecAvg     | 11.9   | 1.1   | 0.9   | 1.2   | 1.2   | 1.4   | 1.0   | 0.6   | 0.6   | 0.9   | 1.0   | 1.1   | 1.1  |
|                                                                                  | PrecMax     | 20.3   | 3.2   | 3.1   | 3.0   | 3.3   | 3.3   | 4.0   | 2.3   | 4.0   | 3.4   | 3.2   | 2.8   | 3.4  |
|                                                                                  | PrecMin     | 5.4    | 0.2   | 0.1   | 0.1   | 0.1   | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1  |
|                                                                                  | PrecStd     | 3.1    | 0.6   | 0.6   | 0.7   | 0.7   | 0.8   | 0.9   | 0.6   | 0.7   | 0.7   | 0.7   | 0.6   | 0.8  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 50.0  | 56.3  | 68.5  | 77.5  | 86.3  | 94.5  | 99.3  | 97.0  | 91.3  | 79.8  | 64.7  | 52.0 |
|                                                                                  |             | MCWB   | 41.4  | 44.0  | 48.1  | 53.4  | 58.1  | 60.6  | 64.1  | 61.5  | 61.1  | 54.6  | 47.7  | 43.0 |
|                                                                                  | 2%          | DB     | 45.2  | 50.2  | 62.8  | 71.3  | 80.9  | 89.5  | 95.9  | 94.0  | 87.2  | 73.5  | 58.9  | 47.7 |
|                                                                                  |             | MCWB   | 38.5  | 41.3  | 46.2  | 50.0  | 56.1  | 60.0  | 62.4  | 60.4  | 58.2  | 52.3  | 45.2  | 40.4 |
|                                                                                  | 5%          | DB     | 41.6  | 46.2  | 57.7  | 66.2  | 76.3  | 85.6  | 93.0  | 91.3  | 83.4  | 68.8  | 54.3  | 43.5 |
|                                                                                  |             | MCWB   | 36.3  | 39.2  | 43.7  | 48.0  | 54.3  | 59.4  | 61.7  | 59.9  | 56.6  | 50.3  | 43.1  | 38.0 |
|                                                                                  | 10%         | DB     | 38.6  | 42.7  | 53.0  | 61.3  | 71.7  | 81.4  | 90.0  | 88.4  | 79.3  | 63.9  | 50.2  | 39.6 |
|                                                                                  |             | MCWB   | 34.6  | 37.3  | 41.9  | 45.9  | 52.6  | 57.8  | 60.9  | 59.3  | 55.2  | 47.8  | 41.4  | 35.6 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 43.0  | 45.2  | 50.0  | 54.2  | 60.3  | 65.0  | 68.7  | 66.1  | 65.0  | 56.5  | 49.8  | 44.0 |
|                                                                                  |             | MCDB   | 48.3  | 53.8  | 63.9  | 75.8  | 79.4  | 87.3  | 88.3  | 87.1  | 80.4  | 75.3  | 59.7  | 50.7 |
|                                                                                  | 2%          | WB     | 39.3  | 42.2  | 47.1  | 51.0  | 57.5  | 62.3  | 66.2  | 63.9  | 61.1  | 54.2  | 46.4  | 41.2 |
|                                                                                  |             | MCDB   | 44.0  | 48.8  | 60.5  | 68.1  | 75.6  | 84.2  | 87.6  | 84.7  | 79.2  | 69.6  | 56.7  | 46.5 |
|                                                                                  | 5%          | WB     | 36.8  | 40.0  | 45.0  | 48.9  | 55.6  | 60.5  | 64.4  | 62.3  | 59.0  | 51.6  | 44.1  | 38.4 |
|                                                                                  |             | MCDB   | 40.9  | 45.5  | 56.0  | 63.8  | 73.1  | 82.0  | 86.1  | 83.9  | 77.0  | 65.7  | 52.9  | 43.0 |
|                                                                                  | 10%         | WB     | 34.8  | 37.4  | 42.6  | 46.8  | 53.6  | 58.7  | 62.8  | 60.8  | 56.8  | 49.3  | 41.9  | 35.6 |
|                                                                                  |             | MCDB   | 38.1  | 42.3  | 52.0  | 59.5  | 69.0  | 78.6  | 84.3  | 82.9  | 75.2  | 62.1  | 49.3  | 39.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 15.1  | 16.5  | 20.9  | 23.8  | 26.6  | 30.3  | 34.8  | 35.2  | 32.3  | 26.5  | 20.6  | 15.0 |
|                                                                                  |             | MCDBR  | 17.4  | 19.9  | 29.3  | 34.5  | 35.9  | 39.2  | 40.6  | 40.7  | 40.9  | 36.2  | 28.2  | 16.8 |
|                                                                                  |             | MCWBR  | 12.1  | 12.8  | 16.3  | 17.1  | 15.7  | 15.9  | 15.3  | 15.4  | 17.2  | 17.9  | 16.7  | 11.3 |
|                                                                                  | 5% WB       | MCDBR  | 15.1  | 18.4  | 26.9  | 30.3  | 32.6  | 36.7  | 35.6  | 34.9  | 33.7  | 31.7  | 25.4  | 15.6 |
|                                                                                  |             | MCWBR  | 10.9  | 12.3  | 15.5  | 15.4  | 14.6  | 15.4  | 14.3  | 13.8  | 14.4  | 16.3  | 15.6  | 10.9 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.246  | 0.257 | 0.270 | 0.296 | 0.312 | 0.314 | 0.330 | 0.328 | 0.302 | 0.278 | 0.257 | 0.245 |      |
|                                                                                  | taud        | 2.477  | 2.531 | 2.584 | 2.495 | 2.498 | 2.522 | 2.487 | 2.471 | 2.544 | 2.595 | 2.590 | 2.534 |      |
|                                                                                  | Ebn at Noon | 292    | 306   | 313   | 309   | 304   | 303   | 297   | 294   | 297   | 294   | 286   | 283   |      |
|                                                                                  | Edh at Noon | 22     | 25    | 27    | 32    | 33    | 33    | 33    | 33    | 28    | 23    | 20    | 19    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 546    | 837   | 1264  | 1690  | 2047  | 2388  | 2386  | 2063  | 1638  | 1088  | 661   | 440   |      |
|                                                                                  | RadStd      | 66     | 99    | 93    | 111   | 191   | 179   | 112   | 102   | 105   | 72    | 57    | 61    |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (12 neighbors) | N/A   | N/A     | N/A    | N/A     | -0.70 | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page